

Cybersecurity Law & Policy Project

NATIONAL CYBERSECURITY LAW FOR VIT'S

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Why VIT's Ecosystem Requires Cybersecurity Law

50 Million

INTERNET USERS NATIONWIDE

DIGITAL LANDSCAPE SCENARIO



Online Banking



E-commerce



Smart Cities



AI Services



Cloud-based Businesses

CRITICAL THREAT MATRIX

— Data Breaches & Ransomware

— Banking Fraud & Identity Theft

— Cyberbullying & Misinformation

CONCLUSION

A national cybersecurity law is necessary to protect citizens, businesses, and critical infrastructure.

Strategic Objectives of the National Law



Key Cyber Law Definitions



Cybercrime

Criminal activities carried out by means of computers or the internet.



Ethical Hacker

A security expert who penetrates systems legally to identify vulnerabilities.



Ransomware

Malicious software that threatens to publish or block access to data unless a ransom is paid.



Data Breach

A security incident in which sensitive data is copied, transmitted, or viewed illegally.



Digital Signature

A mathematical technique used to validate the authenticity and integrity of a message.



Hacker

An individual who uses computers to gain unauthorized access to data.



Malware

Software specifically designed to disrupt, damage, or gain unauthorized access.



Phishing

The practice of sending fraudulent emails to induce individuals to reveal personal info.



Identity Theft

The fraudulent acquisition and use of a person's private identifying information.



Electronic Evidence

Any information or data of investigative value that is stored in electronic form.

Advanced Cyber Law Definitions



Artificial Intelligence

Systems capable of performing tasks that normally require human intelligence.

Importance: Regulates algorithmic bias and automated liability.



Cloud Computing

On-demand availability of computer system resources over the internet.

Importance: Defines jurisdictional boundaries for stored data.



Critical Infrastructure

Physical or virtual assets essential to the functioning of a society.

Importance: Mandates higher security standards for national safety.



Cyberbullying

The use of electronic communication to bully a person, typically by sending messages.

Importance: Criminalizes online harassment and digital stalking.



Deepfake

Synthetic media in which a person in an existing image/video is replaced with someone else.

Importance: Addresses digital impersonation and misinformation fraud.



Encryption

Process of converting information into a code to prevent unauthorized access.

Importance: Safeguards the fundamental right to digital privacy.



Firewall / Zero-Day Exploit

Defense systems vs. unpatched vulnerabilities unknown to the software creator.

Importance: Establishes liability for negligent patch management.



Personal & Sensitive Data

Identifiable information and highly protected categories like health or finance.

Importance: Basis for GDPR/DPDP compliance and breach penalties.

Citizens' Digital Bill of Rights



Right to Privacy

Protection against unauthorized surveillance and data tracking.



Right to Secure Data

Mandatory encryption and protection of personal datasets.



Right to Data Deletion

The "Right to be Forgotten" from corporate databases.



Right to Information

Transparency on how personal data is being processed.



Report Cybercrime

Access to swift reporting portals for digital offenses.



Fair Investigation

Due process and impartial forensic auditing of crimes.



Digital Education

Access to literacy programs for safe internet usage.

GOVERNANCE FRAMEWORK

Responsible Authorities

- Data Protection Board (DPB)
- Cyber Regulatory Authority
- Ministry of Electronics & IT (MeitY)

Enforcement Mechanisms

- Binding Data Audits & Compliance Checks
- Civil Penalties for Breach of Privacy
- Judicial Review of Digital Surveillance

Classification of Cyber Offenses & Crimes



Access & Data Breaches

- Unauthorized Access
- Data Theft
- Website Defacement



Identity & Fraud

- Identity Theft
- Financial Fraud
- Phishing Attacks



Malicious Software

- Ransomware
- Malware Distribution
- Zero-Day Exploits



Social & Information

- Cyberbullying
- Online Harassment
- Misinformation



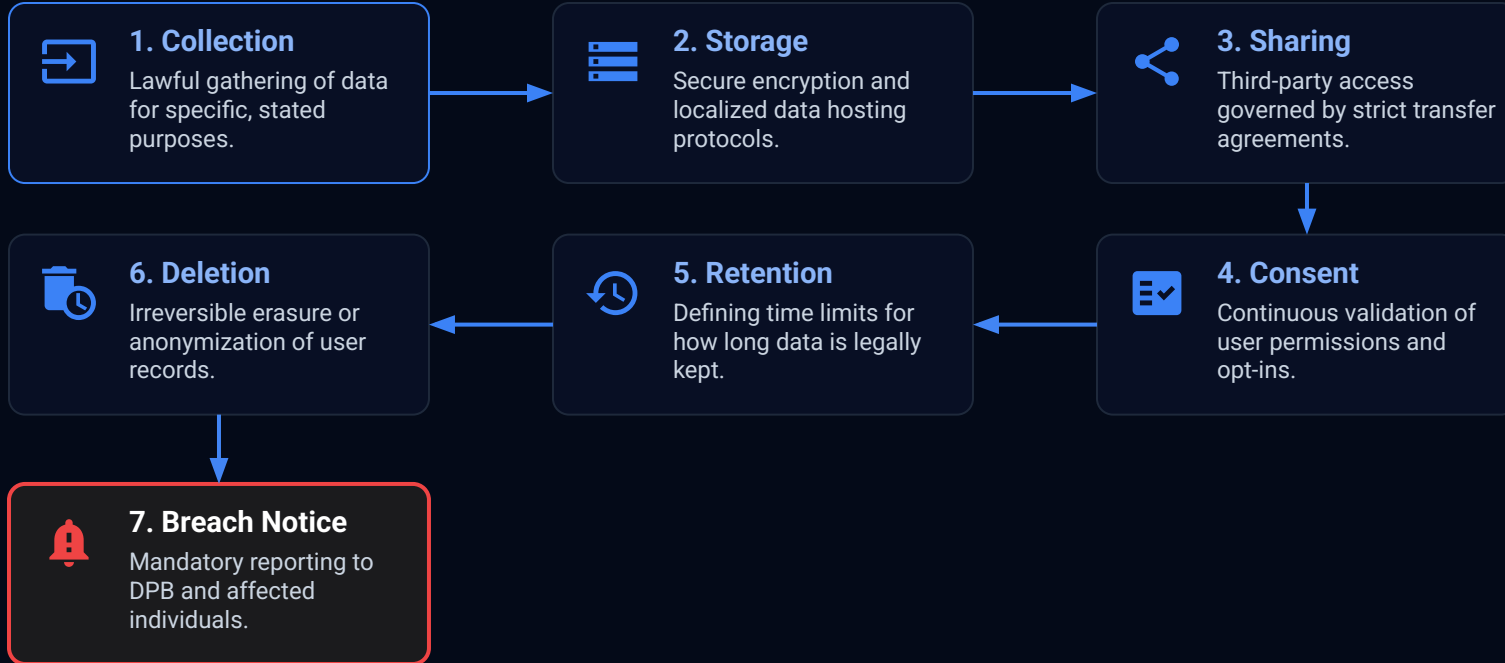
State & Public Safety

- Cyber Terrorism
- Infrastructure Attacks
- Espionage

Legal Punishment & Penalty Framework

Nature of Crime	Max Imprisonment	Fine (Penalty)
Unauthorized Access	Up to 3 Years	₹ 5 Lakh
Data Theft / Breach	Up to 5 Years	₹ 10 Lakh
Identity Theft	Up to 5 Years	₹ 10 Lakh
Phishing Operations	Up to 4 Years	₹ 8 Lakh
Ransomware Attacks	Up to 10 Years	₹ 25 Lakh
Cyber Terrorism	Life Imprisonment	₹ 1 Crore

Data Protection Framework Lifecycle



Data Collection & Storage Compliance Rules

Collection Rules



Lawful & Minimal Gathering

- Data minimization: Collect only what is necessary
- Lawful purpose: Specific, documented business need
- User consent: Explicit opt-in and agreement
- Privacy notices: Transparent disclosure of usage
- Parental consent: Mandatory for processing minor data

Storage Rules



Secure Data Hosting Protocols

- Encryption: Protection for data at rest and in transit
- MFA: Mandatory Multi-Factor Authentication
- Least privilege: Access restricted to essential staff
- Security audits: Regular forensic integrity checks
- Secure backups: Redundant, air-gapped data copies

Data Sharing & Breach Reporting Protocols



Data Sharing Governance

- **Consent Required:** Explicit authorization prior to sharing.
- **Third-party Compliance:** Partners must meet security standards.
- **Cross-border Restrictions:** Limits on international data transfers.
- **Activity Logging:** Immutable audit trails of all sharing events.



Breach Notification Mandates

- **72-Hour Window:** Immediate reporting to regulatory bodies.
- **Citizen Notification:** Informing affected users of exposure.
- **Incident Investigation:** Detailed forensic root-cause analysis.
- **Heavy Penalties:** Fines up to ₹5 Crore for non-compliance.

Corporate Data Responsibilities Matrix



Data Protection

Holistic safeguarding of user information.



Security Audits

Regular systemic integrity verification.



Employee Training

Security awareness and skill building.



Incident Reporting

Mandatory disclosure of anomalies.



Data Officer (DPO)

Appointing specialized legal oversight.



Access Control

Strict identity and permission limits.



Backup & Recovery

Business continuity and data redundancy.



Penetration Testing

Proactive vulnerability identification.



Secure Development

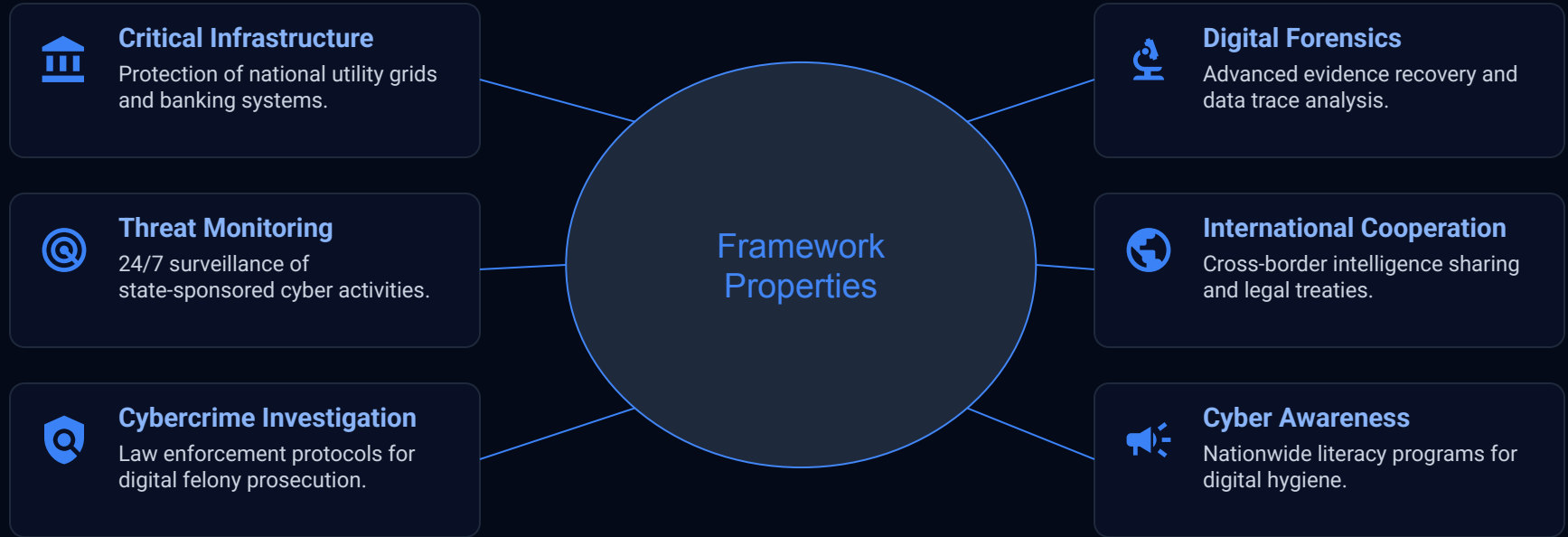
DevSecOps lifecycle integration.



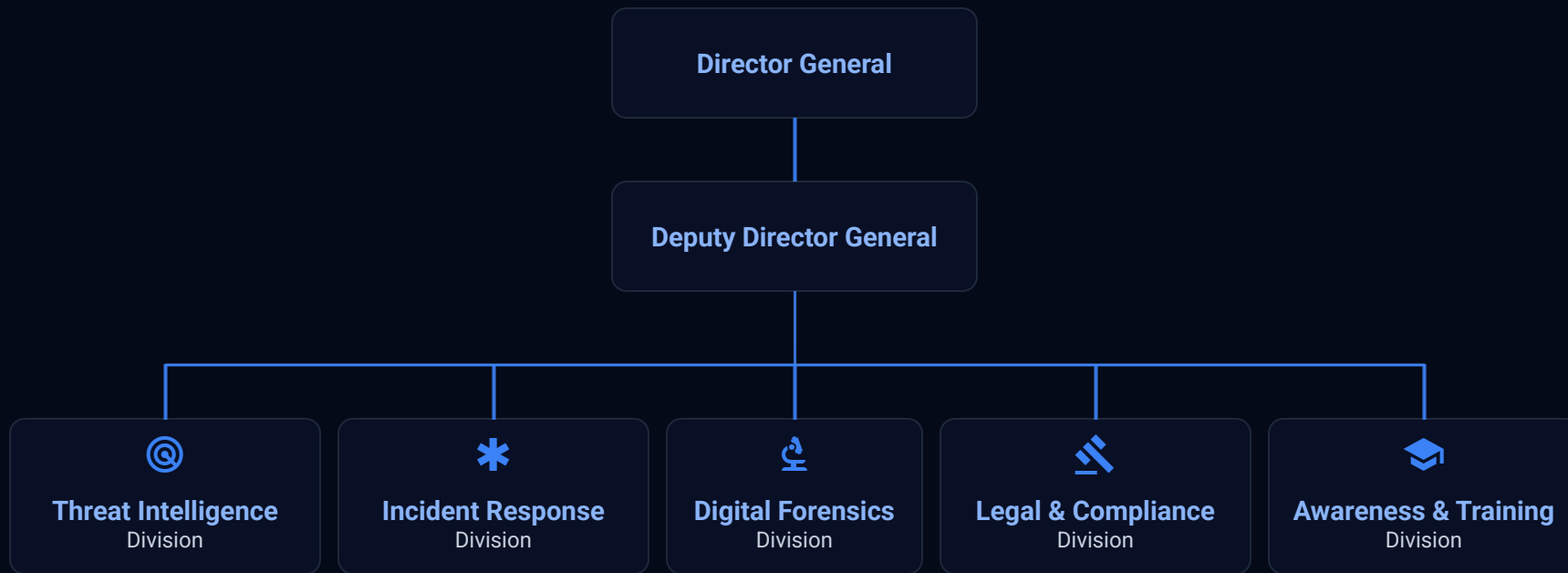
Response Plan

Defined workflow for crisis mitigation.

Government Cyber Security Framework



National Cyber Security Authority (NCSA) Structure



Legal Powers & Regulatory Mandates of NCSA



Cyber Incident Reporting & Mitigation Workflow



International Cyber Security Cooperation



INTERPOL & CERTs

Global law enforcement coordination and emergency response teams.



Sharing Agreements

Bilateral and multilateral pacts for unified digital defense.



Joint Investigations

Cross-border operations to dismantle criminal infrastructure.



Threat Exchange

Real-time intelligence feed on state-sponsored actors.



Extradition Requests

Legal frameworks for prosecuting foreign-based criminals.



Budapest Convention

Alignment with international cybercrime treaty standards.

NCSA Enforcement Case Studies

Case Study 1: Citizen Database Breach



Unauthorized Access & Theft

Critical database breach; citizen PII exfiltrated.



Investigation

Forensic analysis identified weak access controls.



Penalty

Statutory fine and mandatory security overhaul.



Lessons Learned

Zero-trust architecture implementation required.

Case Study 2: National Bank Phishing



Fraud & Losses

Widespread campaign leading to significant financial theft.



Investigation

Coordinated response with Digital Forensics Division.



Punishment

Asset recovery and prosecution under Cyber Act.



Prevention Measures

NCSA mandated bank-wide awareness training.

Case Study 3: National Hospital Ransomware Attack

Attack Intelligence & Investigation



Attack Timeline & Impact

24-hour rapid encryption; Critical patient care systems affected.



Digital Forensics

NCSA identified zero-day exploit in legacy hospital servers.



NCSA Investigation

Trace-back operation across multi-hop VPN servers.



Lessons Learned

Legacy system isolation and offline backup redundancy.

Judicial Outcome & Penalties

Verdicts Delivered

- **Arrest:** Perpetrator apprehended via international CERT cooperation.
- **Imprisonment:** 10-year rigorous sentence under Cyber Act.
- **Financial Fine:** ₹25 Lakh penalty for disruption of public services.

Future of Cybersecurity Policy Roadmap (2025–2035)



Quantum

- Quantum-safe encryption
- PQC standards



Digital Identity

- Secure national digital ID



AI Security

- Regulation of AI systems
- Deepfake controls



Privacy

- Biometric protection
- Behavioral restrictions



Cyber Warfare

- State-sponsored attack defense
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